

Implementation Plan

The Living Map is developed in progressive stages, while preserving the same communication rules: no direct **Writer** → **Command Post**, **Writer** → **Executor**, or **Command Post** → **Executor** mission path outside **ONA Role 4**.

Stage 1: Software Architecture and Dashboard

Period: 5 September 2026 – 5 October 2026

Status: Implemented

The current prototype includes GPS-denied Writer exploration, **fire_detected** and **victim_detected** beacons, BeaconStore, the four ONA roles, satellite/RF mesh communication, Executor selection, operator authorization, and mission-completion feedback.

The validation suite covers normal operation, Writer loss, satellite failure, relay failure, Zombie Link, baseline cases, and combined resilience tests.

Stage 2: Simulation Validation

Period: 5 September 2026 – 5 October 2026

Status: Done

Controlled tests measured beacon preservation, delivery latency, satellite-to-mesh recovery, relay rerouting, heartbeat detection/recovery, packet retries, CRC validation, and mission completion.

Results include approximately **3 s satellite failover**, **less than 1 s rerouting**, approximately **4 s packet recovery after dual-relay failure**, **2 s heartbeat-loss detection**, and **1 s trust recovery**.

Stage 3: Improved Perception

Period: 6 October 2026 – 16 October 2026

Status: Planned

Future work will improve fire/victim detection, confidence fusion, repeated-observation filtering, and localization-error modelling.

All additions must remain compatible with the existing beacon schema, TTL logic, BeaconStore, ONA pipeline, and Command Post workflow.

Stage 4: Physical Beacon and Communication Prototype

Period: 17 October 2026 – 5 November 2026

Status: Planned

The compact beacon format will be transferred to low-power embedded hardware while preserving the same essential fields used in simulation: beacon identifier, event type, local position, timestamp, TTL, and CRC-32 checksum.

For the RF layer, candidate platforms include LoRa transceivers such as the **Semtech SX1262** and **Texas Instruments Sub-1 GHz SimpleLink wireless MCUs** such as the **CC1312R**.

Testing will focus on sub-GHz RF communication, persistent BeaconStore storage, indoor communication range, packet loss, interference, retransmission behavior, beacon lifetime, and deployment reliability.

Stage 5: Physical Writer and Executor Integration

Period: 6 November 2026 – 18 November 2026

Status: Planned

A physical Writer will create confirmed emergency beacons, while a physical Executor will receive authorized missions only through ONA Role 4 and navigate using beacon-derived targets.

Final dispatch remains an operator decision.

Stage 6: Controlled Field Validation

Period: 19 November 2026 – 26 November 2026

Status: Planned

Testing in a controlled emergency environment will measure end-to-end latency, RF range, packet reliability, localization error, Writer-loss preservation, link-failure recovery, mesh rerouting, heartbeat detection, Executor delivery time, and mission success.

Final Integration and Submission

Period: 27 November 2026 – 1 December 2026

Status: Planned

The final period will be used for integration, reliability testing, documentation, demonstration recording, and final submission preparation.

Overall Plan

The plan moves progressively from **validated software to embedded hardware, physical Writer and Executor robots, and controlled field testing**, while preserving the main objective: mission knowledge must remain available and actionable even when the Writer Robot or a communication path fails.

